ASSESSMENT TOOL 1

NAME: Bhuvaneswari S

REG NO:19571048

COURSE NAME: Database Management Systems

COURSE CODE: CSA05

1. UPS Product Tracking Information System (ER Diagram)

Entities

- **Shipped_Item** ***(ItemNo (PK), Weight, Dimensions, InsuranceAmount, Destination, FinalDeliveryDate)***
- **Retail_Center** ***(CenterID (PK), Type, Address)***
- **Transportation_Event** ***(ScheduleNo (PK), Type, DeliveryRoute)***

Relationships

- **Retail Center receives Shipped Item (1:M)**
- **Shipped Item travels through Transportation Event (M:N)**

ER Diagram

```
          Retail Center
 ----------------------------
 CenterID (PK)
 Type
 Address
     |
   receives (1:M)
     |
 ----------------------------
   Shipped Item
 ----------------------------
 ItemNo (PK)
 Weight
 Dimensions
 InsuranceAmount
 Destination
 FinalDeliveryDate
     |
  travels through (M:N)
     |
```

Transportation Event

ScheduleNo (PK)

Type

DeliveryRoute

---

2. Course Management System

Entities

- **Course**
- **Section (Weak Entity)**
- **Room**
- **Exam**

Relationships

Course

(Name, Dept, CourseNo)

|

has

|

Section (Weak)

(SecNo, Enrollment)

|

scheduled in

|

Exam

(Time)

|

conducted in

|

Room

(RoomNo, Capacity, Building)

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3. University Registrar Office ER Diagram

Entities

- **Student**
- **Course**
- **Course Offering**
- **Instructor**

Relationships

Student

(StudentID, Name, Program)

|

Enrolls

|

Course Offering

(Year, Semester,

Section, Timing)

|

belongs to

|

Course

(CourseNo, Title,

Credits, Syllabus)

|

taught by

|

Instructor

(ID, Name,

Department, Title)

Enrollment

contains Grade

---

4. Hospital Management System

Entities

- **Patient**
- **Physician**
- **Test Log**
- **Patient History**

ER Diagram

Patient

(SS#, Name, Insurance)

|

treated by

|

Physician

(Name, Specialization)

Patient

|

undergoes

|

Test Log

(TestName, Date, Time)

Test Log

|

stored in

|

Patient History

Tables

Patient

| SS# | Name | Insurance |

Physician

| Name | Specialization |

Test_Log

| SS# | Test Name | Date | Time |

Doctor_Patient

| Physician | SS# |

Patient_History

| SS# | Test Name | Date |

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5. Company Project Management System

Department

(DeptNo, DeptName)

|

controls

|

Project

(ProjectNo, Name,

Location)

Department

|

employs

|

Employee

(SSN, Name,

Address,

Salary,

Sex,

BirthDate)

Employee

|

works on

|

Project

(Hours/Week)

Employee

|

supervises

|

Employee

Employee

|

has

|

Dependent

(Name,

BirthDate,

Relationship)

---

6. Schema and Instance

Schema

A schema is the blueprint or structure of the database.

Example

Student(StudentID, Name, Department)

This structure rarely changes.

Instance

An instance is the actual data stored in the database at a particular time.

Example

StudentID	Name	Department
101	Anitha	CSE
102	Rahul	ECE

Types of Schema

Logical Schema	Physical Schema	View Schema
Defines tables and relationships	Defines storage details	Defines user-specific views
Logical design	Physical storage	External view
Used by designers	Used by DBMS	Used by end users

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7. Bank Management System

Entities

- **Customer**
- **Account**
- **Branch**
- **Loan**

Relationships

Customer

(CustomerID,

Name,

Phone)

|

owns

|

Account

(AccountNo,

Type,

Balance)

|

belongs to

|

Branch

(BranchID,

Name,

Location)

Customer

|

takes

|

Loan

(LoanID,

Amount)

---

8. Extended ER (EER) Model

The Enhanced Entity Relationship (EER) model extends the ER model by introducing additional concepts.

Generalization

Combines multiple lower-level entities into one higher-level entity.

Example

Car

Bike

↓

Vehicle

Specialization

Divides one entity into specialized subtypes.

Employee

↓

Manager

Developer

Tester

Aggregation

Treats a relationship as a higher-level entity.

Employee

|

Works On

|

Project

↓

Assignment

---

9. Hospital Management System (EER Diagram)

Person

|

| |

Patient Doctor

Doctor

|

treats

|

Patient

Patient

|

has

|

Medical Record

(Weak Entity)

Medical Record

depends on Patient

Generalization

Person

↓

Doctor

Patient

Weak Entity

Medical Record

---

10. Database Administrator (DBA)

A Database Administrator (DBA) is responsible for installing, managing, securing, and maintaining the database system.

Responsibilities

- **Install and configure DBMS**
- **Create databases**
- **Define users and permissions**

- **Ensure database security**
- **Backup and recovery**
- **Monitor performance**
- **Optimize queries**
- **Maintain data integrity**
- **Handle concurrency control**
- **Manage storage and indexing**